Jize Jiang

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Advisor: ChengXiang Zhai

Research Interests

Evaluation and diagnostics of logical reasoning in LLMs, **memory-augmented LLM agents**, **post-training scaling for reasoning and tool use in LLMs/VLMs**, **human-centered tools for understanding and steering AI agents**.

Education

- **University of Illinois Urbana-Champaign** — Champaign, IL
  Ph.D. in Computer Science; Advisor: ChengXiang Zhai — *2025 – Present*
- **University of Illinois Urbana-Champaign** — Champaign, IL
  M.S. in Computer Science; Thesis: Diagnostic Evaluation of Logical Reasoning in LLMs — *2023 – 2025*
- **University of Illinois Urbana-Champaign** — Champaign, IL
  B.S. in Mathematics & Computer Science and Statistics — *2018 – 2023*

Publications

$*$ *indicates equal contribution.*

Conference Papers

[1] ***VGI-Bench: Probing Visual Intelligence in Video Generation Models***.
Xuan He, Cong Wei, Yuhao Cheng, Linrui Ma, Yuxuan Zhang, Zuojun Li, Yuhao Wen, **Jize Jiang**, Zeyi Liu, Yuren Hao, Songcheng Cai, Keming Wu, Penghui Du, Kai Zou, Rui Yang, Chenkai Sun, Ke Yang, Ping Nie, Kelsey R. Allen, Chenglong Wang, Michel Galley, Jianfeng Gao, ChengXiang Zhai.
EMNLP 2026. [arXiv]

[2] ***PlugMem: A Task-Agnostic Plugin Memory Module for LLM Agents***.
Jize Jiang*, Ke Yang*, Zixi Chen*, Xuan He*, Michel Galley, Chenglong Wang, Jianfeng Gao, Jiawei Han, ChengXiang Zhai.
ICML 2026. [arXiv] [Code]

[3] ***VTool-R1: VLMs Learn to Think with Images via Reinforcement Learning on Multimodal Tool Use***.
Mingyuan Wu*, Jingcheng Yang*, **Jize Jiang**, Meitang Li, Kaizhuo Yan, Hanchao Yu, Minjia Zhang, ChengXiang Zhai, Klara Nahrstedt.
ICLR 2026. [arXiv] [Code]

[4] ***Cache-of-Thought: Master-Apprentice Framework for Cost-Effective Vision Language Model Inference***.
Jize Jiang*, Mingyuan Wu*, Haozhen Zheng*, Meitang Li, Zhaoheng Li, Beitong Tian, Bo Chen, Yongjoo Park, Minjia Zhang, ChengXiang Zhai, Klara Nahrstedt.
EMNLP 2025. [Paper] [Code]

[5] ***UOUO: Uncontextualized Uncommon Objects for Measuring Knowledge Horizons of Vision Language Models***.
Jize Jiang*, Xinyu Pi*, Mingyuan Wu*, Haozhen Zheng*, Beitong Tian, ChengXiang Zhai, Klara Nahrstedt, Zhiting Hu.
EMNLP 2024. [Paper] [Code]

[6] ***Competence-Based Analysis of Language Models***.
Adam Davies, **Jize Jiang**, ChengXiang Zhai.
NeurIPS 2024 Workshop on Causality and Large Models (CaLM). *Oral Presentation.* [arXiv]

Preprints & Under Review

[7] ***HarnessLens: Interactive Visualization for Understanding and Evaluating Unfamiliar Agentic AI Systems***.
Zepei Li*, Dina Albassam*, **Jize Jiang***, ChengXiang Zhai, Yun Huang.
Under review (CHI 2027).

[8] ***Synthetic WideSearch: Novel Benchmark and Baselines***.
Dean Alvarez, **Jize Jiang**, Joshua Wong, Rebecca Wong, ChengXiang Zhai.
Preprint, 2026.

[9] ***SVR-R1: Bootstrapping Multi-modal Reasoning with Self-verification in Reinforcement Learning***.
Mingyuan Wu, Jingcheng Yang, Shengyi Qian, Xudong Wang, **Jize Jiang**, Qifan Wang, Aashu Singh, Khoi Pham, Fei Liu, Zhaolun Su, Zhuokai Zhao, Klara Nahrstedt, Jianyu Wang, Hanchao Yu.
Preprint, 2026. [arXiv]

[10] ***Competence-Based Evaluation of Logical Reasoning in Large Language Models***.
Jize Jiang, ChengXiang Zhai.
In submission to ACL Rolling Review (ARR). [arXiv]

[11] ***Spreadsheet-RL: Advancing Large Language Model Agents on Realistic Spreadsheet Tasks via Reinforcement Learning***.
Banghao Chi, Yining Xie, Mingyuan Wu, Jingcheng Yang, **Jize Jiang**, Zhaoheng Li, Shengyi Qian, Minjia Zhang, Klara Nahrstedt, Rui Hou, Xiangjun Fan, Hanchao Yu.
Under review (ICLR 2027). [arXiv]

[12] ***Aha Moment Revisited: Are VLMs Truly Capable of Self-Verification in Inference-time Scaling?***
Mingyuan Wu*, Meitang Li*, Jingcheng Yang*, **Jize Jiang**, Kaizhuo Yan, Zhaoheng Li, Hanchao Yu, Minjia Zhang, Klara Nahrstedt.
Under review, 2025. [arXiv]

[13] ***Efficient Title Reranker for Fast and Improved Knowledge-Intense NLP***.
Ziyi Chen, **Jize Jiang**, Daqian Zuo, Heyi Tang, Jun Yang, Yuxiang Wu, Kevin Chang.
Preprint, 2023. [arXiv]

Work in Progress

[14] ***PlugMem Studio: An Interactive Memory Inspection Interface for Agentic Systems***.
In preparation.

[15] ***Language-Conditioned Affordances for Grasping in a Cluttered Environment***.
In preparation.

Research Experience

- **TIMAN Research Group, University of Illinois Urbana-Champaign** — Champaign, IL
  Graduate Research Assistant; Advisor: ChengXiang Zhai — *2022 – Present*
  - **PlugMem**: Designed a task-agnostic plugin memory module attaching to arbitrary LLM agents without task-specific redesign, structuring raw interactions into a memory graph of propositional facts and prescriptive skills with intent-aware retrieval. Across three heterogeneous benchmarks (long-horizon conversational QA, multi-hop knowledge retrieval, web agent tasks), PlugMem unchanged outperformed both task-agnostic and task-specific baselines while achieving highest information density. In collaboration with **Microsoft Research** (**ICML 2026**).
  - **VGI-Bench**: Contributed to a benchmark of 27 tasks and 810 instances probing whether video generation models perform visually grounded reasoning rather than surface-level synthesis; the strongest model reached only 51% accuracy, with analysis showing later denoising steps refine early hypotheses instead of correcting reasoning errors (**EMNLP 2026**).
  - **VTool-R1**: Contributed to the first framework training VLMs to produce multimodal chains of thought—interleaving text with intermediate visual reasoning steps via Python tool invocations—using outcome-based RL without process supervision, achieving 10%+ accuracy gains on chart/table QA (**ICLR 2026**).
  - **Cache-of-Thought**: Designed a master-apprentice framework for cost-effective VLM reasoning, where a large master VLM generates high-quality cached QA pairs retrieved via dual-modality dense retrieval to boost small apprentice VLMs through in-context learning. Achieved up to 7.7% overall performance gain under fixed budget and up to 36.6% apprentice VLM improvement (**EMNLP 2025**).
  - **HarnessLens**: Co-led an interactive visualization system that helps practitioners understand and evaluate agentic AI systems they did not build, surfacing execution traces and harness behavior for inspection (under review, **CHI 2027**).
  - **Aha Moment Revisited**: Systematically evaluated self-verification capabilities of RL-tuned VLMs during inference-time scaling.

- **M.S. Thesis – Diagnostic Evaluation of Logical Reasoning in LLMs**: Designed a diagnostic framework using self-generating synthetic ordering tasks with controlled complexity, formulating problems as two-layer representations (abstract logical structures + natural language surface forms) with controlled perturbations (equivalent rephrasings, directional query variations) to test reasoning invariance. Revealed significant linguistic sensitivity and pronounced answer-type biases in GPT-4.1 and GPT-4o, indicating reliance on heuristic shortcuts over genuine logical deduction. Extended into ***Competence-Based Evaluation of Logical Reasoning in Large Language Models*** (in submission to ACL Rolling Review).

- **Automatic Reasoning Lab, Langboat Inc.** Remote
  Research Intern; Supervised by Dr. Ming Zhou *Jan 2022 – May 2022*
  - Surveyed and benchmarked approaches for low-resource semantic parsing and automatic reasoning.
  - Built a neural-symbolic reasoning system for the **AR-LSAT** (Analytical Reasoning section of the Law School Admission Test) task, enabling integration of explicit logical rules with end-to-end differentiable training, outperforming baselines on ordering-related subtasks.

- **Microsoft Research Asia, System Intelligence Group** Beijing, China
  Research Intern; Mentor: Yaming Yang *Aug 2021 – Dec 2021*
  - Researched transferable graph neural networks, developing ego-graph encoding techniques to produce richer node embeddings for improved cross-domain generalization.
  - Designed and evaluated structural encoding methods capturing local neighborhood topology to enhance node representation learning in heterogeneous graph settings.

- **Sanmi Koyejo Lab, University of Illinois Urbana-Champaign** Champaign, IL
  Undergraduate Research Assistant; Mentor: Brando Miranda *Jun 2020 – Aug 2023*
  - **IsabelleGym**, a training and testing environment for automated theorem proving agents, translating Isabelle's proof archives into Python-friendly datasets and integrating Isabelle (Scala/ML) into a machine-friendly Python package.
  - **Meta-Learning to Prove via Complexity Minimization**: meta-learned a position evaluation function (estimating remaining proof steps) as a loss function to improve tactic prediction and generate shorter proofs.
  - **Ultimate-pycoq**, an ML interface for the Coq theorem prover; for the first time in the Coq community, collected and processed an archive of proofs with proof steps and states.

- **Illinois Risk Lab, University of Illinois Urbana-Champaign** Champaign, IL
  Undergraduate Research Assistant; Supervisor: Dr. Zhiyu Frank Quan *Mar 2021 – Dec 2022*
  - Performed data cleaning and preprocessing on 500K business and insurance records with high dimensionality and mixed data types.
  - Built end-to-end keyword extraction and text classification models for risk analysis, replacing the original rule-based pipeline.
  - Developed a Question Answering system for risk-related queries over text and numerical sources.

Professional Service

- Reviewer: **ICML 2026**

Teaching Experience

- Teaching Assistant: **CS 357** Numerical Methods I, **CS 450** Numerical Analysis, **CS 447** Natural Language Processing
- Course Assistant: **CS 222** Software Design Lab, **CS 173** Discrete Structures, **CS 125** Introduction to Computer Science